Lab Experiment: 2

Title: Dataset Creation and Data Warehouse Star Model Design

*Course title: **Data Mining Problem Description***

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Project Title: *Intelligent Thesis Supervisor Recommendation and Performance Prediction System (ThesisHub)*

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1. Introduction

Universities generate large volumes of academic data related to students, teachers, thesis supervision, and evaluation processes. However, this data is often stored in isolated operational databases that are not optimized for analytical processing. For data mining and intelligent decision-making, it is necessary to design a structured dataset and a data warehouse model that supports efficient querying and pattern analysis.

In this experiment, a dataset is created from the ThesisHub system and a **Star Schema Data Warehouse Model** is designed to support data mining tasks such as supervisor recommendation, student performance prediction, and academic trend analysis. The star model provides a simplified and optimized structure for analytical queries.

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2. Objective of the Experiment

The main objectives of this experiment are:

- To design a structured dataset for ThesisHub data mining operations.
- To identify fact and dimension tables for the academic data warehouse.
- To design a star schema model suitable for analytical processing.
- To prepare the dataset for further mining and machine learning experiments.

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3. Dataset Creation

The dataset is extracted from the ThesisHub system database which stores academic and activity-related information. The collected dataset consists of both **transactional data** and **descriptive data**.

3.1 Data Sources

The dataset is collected from the following system modules:

- Student Management Module
- Teacher Management Module
- Thesis Supervision Module
- Task Evaluation Module
- System Activity Logs

3.2 Dataset Attributes

The dataset includes the following attributes:

Student Data

- Student_ID
- CGPA
- Department
- Semester
- Attendance_Percentage

Student_ID	CGPA	Department	Semester	Attendance_Percentage
S101	3.85	CSE	8	92
S102	3.20	CSE	8	85
S103	3.60	CSE	8	88
S104	2.95	CSE	8	76
S105	3.75	CSE	8	90
S106	3.10	CSE	8	82

Teacher Data

- Teacher_ID
- Expertise_Area
- Supervised_Group_Count
- Past_Evaluation_Rating

Teacher_ID	Expertise_Area	Supervised_Group_Count	Past_Evaluation_Rating
T201	Artificial Intelligence	2	4.7
T202	Data Science	1	4.4
T203	Networking	2	4.1
T204	Software Engineering	1	4.5
T205	Cyber Security	1	

Thesis Data

- Thesis_ID
- Thesis_Topic
- Submission_Duration
- Final_Marks
- Supervisor_Feedback_Score

Thesis_ID	Thesis_Topic	Submission_Duration (Days)	Final_Marks	Supervisor_Feedback_Score
TH301	Smart Attendance System	110	88	4.8
TH302	Student Result Prediction System	130	75	4.1
TH303	Network Security Framework	150	65	3.8
TH304	Thesis Management System	105	90	4.9
TH305	Face Recognition Attendance	115	82	4.6

System Activity Data

- Login_Frequency
- Task_Submission_Count
- Progress_Update_Frequency

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4. Data Preprocessing

Before storing the dataset in the data warehouse, preprocessing is performed to ensure data quality and consistency.

4.1 Data Cleaning

- Duplicate records are removed.
- Missing CGPA and evaluation values are handled using average value substitution.
- Invalid entries are filtered out.

4.2 Data Normalization

- CGPA values are normalized between 0 and 1.
- Marks and scores are scaled to maintain uniform ranges.

4.3 Feature Selection

Only relevant attributes are selected to improve mining efficiency:

- CGPA
- Submission duration
- Final marks
- Supervisor feedback score
- Attendance percentage

4.4 Data Encoding

Categorical attributes such as department and expertise area are converted into numerical format using encoding techniques.

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5. Data Warehouse Design Approach

A **Star Schema Model** is selected for the ThesisHub data warehouse due to the following advantages:

- Simple structure
- Faster query performance
- Easy integration with OLAP tools
- Efficient support for aggregation and reporting

The star schema consists of one central **Fact Table** connected to multiple **Dimension Tables**.

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6. Star Schema Design for ThesisHub

6.1 Fact Table: Thesis_Performance_Fact

This table stores measurable and analytical data.

Attributes:

- Fact_ID (Primary Key)
- Student_ID (Foreign Key)
- Teacher_ID (Foreign Key)
- Thesis_ID (Foreign Key)
- Time_ID (Foreign Key)
- Final_Marks
- Submission_Duration
- Supervisor_Feedback_Score
- Task_Completion_Rate

Fact_ID	Student_ID	Teacher_ID	Thesis_ID	Time_ID	Final_Marks	Submission_Days	Feedback_Score	Task_Completion_%
F001	S101	T201	TH301	TM01	88	110	4.8	95
F002	S102	T202	TH302	TM01	75	130	4.1	85
F003	S103	T201	TH301	TM02	82	115	4.6	90
F004	S104	T203	TH303	TM02	65	150	3.8	70
F005	S105	T204	TH304	TM03	90	105	4.9	97

6.2 Dimension Tables

Student_Dimension

- Student_ID (Primary Key)
- CGPA
- Department
- Semester
- Attendance_Percentage

Student_ID	CGPA	Department	Semester	Attendance_%
S101	3.85	CSE	8	92
S102	3.20	CSE	8	85
S103	3.60	CSE	8	88

S104	2.95	CSE	8	76
S105	3.75	CSE	8	90

Teacher_Dimension

- Teacher_ID (Primary Key)
- Expertise_Area
- Supervised_Group_Count
- Evaluation_Rating

Teacher_ID	Expertise_Area	Supervised_Groups	Evaluation_Rating
T201	AI	2	4.7
T202	Data Science	1	4.4
T203	Networking	2	4.1
T204	Software Engineering	1	4.5

Thesis_Dimension

- Thesis_ID (Primary Key)
- Thesis_Topic
- Research_Domain
- Thesis_Type

Thesis_ID	Thesis_Topic	Research_Domain	Thesis_Type
TH301	Smart Attendance System	AI	Application
TH302	Student Result Prediction	Data Science	Research
TH303	Network Security Model	Networking	Research
TH304	Thesis Management System	Software Engineering	Application

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Time_Dimension

- Time_ID (Primary Key)
- Semester
- Year
- Month

Time_ID	Month	Semester	Year

TM01	January	Spring	2025
TM02	June	Summer	2025
TM03	October	Fall	2025

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7. Star Schema Structure (Text Sketch)

```
                       Student_Dimension
                     ---------------------
                    | Student_ID (PK)      |
                    | CGPA                 |
                    | Department           |
                    | Semester             |
                    | Attendance_%         |
                     ---------------------
                             |
                             |
                             v
Teacher_Dimension ----> Thesis_Performance_Fact <---- Thesis_Dimension
-----------------      ------------------------      ------------------
Teacher_ID (PK)         Fact_ID (PK)                  Thesis_ID (PK)
Expertise_Area          Student_ID (FK)               Thesis_Topic
Evaluation_Rating       Teacher_ID (FK)               Research_Domain
Supervised_Groups       Thesis_ID (FK)                Thesis_Type
                         Time_ID (FK)
                         Final_Marks
                         Submission_Days
                         Feedback_Score
                         Task_Completion_%
                             |
                             |
                             v
                        Time_Dimension
                     ---------------------
                    | Time_ID (PK)         |
                    | Month                |
                    | Semester             |
                    | Year                 |
                     ---------------------
```

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8. Benefits of Star Model for ThesisHub

The star schema provides the following advantages:

- Faster analytical query processing
- Simplified joins between fact and dimension tables
- Efficient support for aggregation queries
- Easy expansion for future attributes
- Improved performance for mining algorithms

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9. Application of Star Schema in Data Mining

The designed star model supports the following mining operations:

Supervisor Recommendation

- Query student performance and teacher success rate
- Match suitable supervisors based on expertise and outcomes

Performance Prediction

- Extract historical student and thesis performance records
- Train classification and regression models

Student Clustering

- Group students based on CGPA, attendance, and submission behavior

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10. Expected Outcome

After implementing the dataset and star schema design, the following outcomes are expected:

- Clean and structured dataset ready for mining
- Efficient data retrieval for analytics
- Improved performance of machine learning algorithms
- Better academic decision support

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11. Conclusion

This experiment successfully designs a dataset and star schema data warehouse model for the ThesisHub system. The structured data warehouse enables efficient data analysis, pattern discovery, and predictive modeling. The star schema approach simplifies analytical processing and provides a strong foundation for implementing intelligent recommendation and performance prediction systems in the academic domain.
